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Senior Software Engineer — Java · Distributed Systems · AWS

BEING AN ACCOUNT OF THREE SEASONS AT AMAZON TRAVEL

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Wherein are recounted the principal achievements of a Java-first backend engineer presently shipping pricing, checkout, and payment infrastructure for *Amazon Travel* — hotels, flights, and insurance. Partial to *CompletableFuture*, idempotent webhooks, small blast radii, and runbooks written before the incident. Three seasons completed; a fourth solicited.

P90 LATENCY	CONVERSION	COST / TXN	ON-CALL
−47%	**+33%**	**−43BPS**	**20+ → 5**
1.10s → 0.59s on pricing fan-out	*traveler booking funnel*	*idempotent payment pipeline*	*pages / shift · 150+ incidents resolved*

Chapter the First

SDE II — Amazon Travel

AMAZON TRAVEL — *Hotels · Flights · Insurance*

JUL 2023 – SEP 2025 · BENGALURU

Owner of four Java microservices across the booking funnel — pricing, checkout, payment webhooks, insurance attach.

I. *On the matter of latency.* Re-architected pricing fan-out with `CompletableFuture.allOf` and bulkhead `ExecutorService` pools, isolating slow suppliers from the hot path. **P90 1.10s → 0.59s (−47%)**; traveler conversion **+33%**.

II. *Of payments & idempotence.* Idempotent payment webhook on `SQS + Step Functions` with DynamoDB dedupe keys — cost-per-transaction **−43 bps** and the double-capture incident class extinguished.

III. *A matter of infrastructure.* Rewrote six legacy pipelines into `Java CDK` with versioned stacks and blue/green cutovers; rollback rate **halved** and MTTR reduced by **one day**.

IV. *Of stewardship.* Pruned on-call load from **20+ to ~5 pages per shift**; config-hygiene violations **5–6 → 0**; resolved **150+ incidents**; stopped an estimated **~30 bps** funnel-churn on insurance attach.

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Chapter the Second

Previous Chapters

SDE I — Amazon Travel
JUL 2022 – JUL 2023 · BENGALURU

Built REST endpoints and `@Repository` DynamoDB DAOs wired with **Spring DI**; shipped auto-insurance renewal contributing **+60% renewal completion**. Held the line at **80%+ JUnit coverage** with **JaCoCo** enforced in CI.

SDE Intern — Amazon Travel
MAY 2021 – FEB 2022 · BENGALURU

Led the **GraphQL → Java** migration of a checkout surface; delivered the One-Click Renewal flow with `Guice` DI and an end-to-end harness at **90%+ coverage**.

❦ THE APPARATUS ❦

A.	*JVM Core*	Java 17 · Spring Boot · Spring MVC · Spring DI · Guice · Maven · Gradle
B.	*Concurrency*	ExecutorService · CompletableFuture · ConcurrentHashMap · ScheduledExecutor · bulkheads
C.	*AWS*	Lambda · DynamoDB · SQS · Step Functions · API Gateway · AppSync · CloudWatch · ECS · CDK (Java)
D.	*Quality*	JUnit 5 · Mockito · JaCoCo · integration harnesses · contract tests
E.	*Front-End*	React (SSR & CSR) · Redux · TypeScript · Vitest
F.	*Data Patterns*	DynamoDB single-table · idempotency keys · dedupe queues
G.	*Ops*	runbooks · alarm hygiene · SEV pages · on-call rotations
H.	*Method*	trunk-based · code-review discipline · postmortems · SLO-first

❧ SUNDRY DEVICES ❧

Redis Server — in Java
github.com/praxstack/redis-server-java

From-scratch **RESP2 protocol** with event-loop I/O and a hashmap engine. Written for sport & study; benchmarked against the C implementation to keep one honest about what the JVM can actually do.

`SET 154k rps` `GET 195k rps` `INCR 128k rps` `42 tests` `~78% of C`

Markdown Viewer App
github.com/praxstack/markdown-viewer-app

Zero-config Markdown reader with a plugin architecture and a keyboard-first command palette. Built to fit in a browser tab and stay there — dependency-free at runtime, tested obsessively.

`155+ Vitest tests` `<100KB gzipped` `0 runtime deps`

Open-Source Contributions
Open-source: 3 PRs merged into danielmiessler/Fabric (42k-star Go AI-tooling framework) — Bedrock bearer-token auth, dynamic region fetching, and a streaming-deadlock fix.

❦ LETTERS OF REFERENCE ❦

of formal instruction

M.C.A. — Computer Science · MNNIT · Prayagraj — 2019–2022 · GPA 9.25/10

B.Sc. (Hons) — Computer Science · Bhaskaracharya College · Delhi University — 2015–2018

of recognitions received

★ **Amazon Pay Merchants** — Categories Champion, 2023

★ **100+ trust-buster bugs** resolved across travel-funnel quality sprints